

Chapter 4

Carbon and Its Compounds

Chapter 4 – Carbon and Its Compounds (30 MCQs with Answers)

1. The valency of carbon is:

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C) 4

2. The ability of carbon to form long chains with itself is called:

- A) Isomerism
- B) Catenation
- C) Crystallisation
- D) Hydrogenation

Answer: B) Catenation

3. Which of the following is the molecular formula of methane?

- A) CH_3
- B) CH_4
- C) C_2H_6
- D) C_3H_8

Answer: B) CH_4

4. Which of the following is the general formula of alkanes?

- A) C_nH_{2n}
- B) $\text{C}_n\text{H}_{2n+2}$
- C) $\text{C}_n\text{H}_{2n-2}$
- D) $\text{C}_n\text{H}_{2n-4}$

Answer: B) $\text{C}_n\text{H}_{2n+2}$

5. The compound with a triple bond between two carbon atoms is:

- A) Ethane
- B) Ethene
- C) Ethyne
- D) Propane

Answer: C) Ethyne

6. The functional group -OH represents:

- A) Aldehyde
- B) Alcohol
- C) Ketone
- D) Carboxylic acid

Answer: B) Alcohol

7. The compound CH_3COOH is:

- A) Ethanol
- B) Acetic acid
- C) Methanoic acid
- D) Acetone

Answer: B) Acetic acid

8. The process of converting vegetable oils into fats using hydrogen is called:

- A) Hydrolysis
 - B) Hydrogenation
 - C) Oxidation
 - D) Saponification
- Answer: B) Hydrogenation
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9. Which metal is used as a catalyst in hydrogenation?

- A) Iron
- B) Nickel
- C) Zinc
- D) Copper

Answer: B) Nickel

10. The molecular formula of ethene is:

- A) C_2H_4
- B) C_2H_2
- C) CH_4
- D) C_3H_6

Answer: A) C_2H_4

11. The molecular formula of propene is:

- A) C_3H_8

B) C_3H_6

C) C_3H_4

D) C_4H_8

Answer: B) C_3H_6

12. The simplest hydrocarbon is:

A) Methane

B) Ethane

C) Ethene

D) Ethyne

Answer: A) Methane

13. Which of the following is a saturated hydrocarbon?

A) Ethene

B) Ethyne

C) Ethane

D) Propene

Answer: C) Ethane

14. Unsaturated hydrocarbons contain:

A) Single bonds only

B) Double or triple bonds

C) Only triple bonds

D) No bonds

Answer: B) Double or triple bonds

15. Which of the following statements is true about covalent compounds?

A) They have high melting points

B) They are good conductors

C) They have low melting and boiling points

D) They are ionic in nature

Answer: C) They have low melting and boiling points

16. The reaction between an acid and an alcohol to form an ester is called:

A) Esterification

B) Saponification

C) Hydrogenation

D) Neutralisation

Answer: A) Esterification

17. The sweet-smelling compound formed by esterification is:

- A) Soap
- B) Ester
- C) Alcohol
- D) Acid

Answer: B) Ester

18. The compound used in alcoholic drinks is:

- A) Methanol
- B) Ethanol
- C) Ethanoic acid
- D) Propanol

Answer: B) Ethanol

19. The harmful alcohol which causes blindness is:

- A) Methanol
- B) Ethanol
- C) Propanol
- D) Butanol

Answer: A) Methanol

20. The process of making soap from fat and alkali is called:

- A) Esterification
- B) Neutralisation
- C) Saponification
- D) Hydrogenation

Answer: C) Saponification

21. The functional group present in aldehydes is:

- A) -COOH
- B) -CHO
- C) -OH
- D) -CO-

Answer: B) -CHO

22. Which of the following compounds belongs to the alcohol group?

- A) CH_3OH

- B) CH_3COOH
 - C) HCHO
 - D) C_2H_2
- Answer: A) CH_3OH
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23. The general formula for alkynes is:

- A) $\text{C}_n\text{H}_{2n+2}$
- B) C_nH_{2n}
- C) $\text{C}_n\text{H}_{2n-2}$
- D) $\text{C}_n\text{H}_{2n-4}$

Answer: C) $\text{C}_n\text{H}_{2n-2}$

24. Carbon forms covalent bonds by:

- A) Gaining electrons
- B) Losing electrons
- C) Sharing electrons
- D) Transferring protons

Answer: C) Sharing electrons

25. Which gas is evolved when sodium carbonate reacts with ethanoic acid?

- A) Hydrogen
- B) Carbon dioxide
- C) Oxygen
- D) Methane

Answer: B) Carbon dioxide

26. The common name of ethanoic acid is:

- A) Formic acid
- B) Acetic acid
- C) Citric acid
- D) Oxalic acid

Answer: B) Acetic acid

27. The number of covalent bonds in methane is:

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C) 4

28. Which of the following is an organic acid?

A) HCl

B) H_2SO_4

C) CH_3COOH

D) HNO_3

Answer: C) CH_3COOH

29. Ethanol reacts with sodium to form:

A) Hydrogen gas

B) Oxygen gas

C) Carbon dioxide

D) Chlorine gas

Answer: A) Hydrogen gas

30. The IUPAC name of $\text{CH}_3\text{CH}_2\text{OH}$ is:

A) Methanol

B) Ethanol

C) Propanol

D) Butanol

Answer: B) Ethanol
